

Problem

Determine the voltage input to the inverting terminal of an op amp when $-40\ \mu\text{V}$ is applied to the noninverting terminal and the output through an open-loop gain of 150,000 is 15 V.

Step-by-step solution

Step 1 of 2

Write the expression for output voltage of op-amp.

$$\begin{aligned} v_o &= Av_d \\ &= A(v_2 - v_1) \end{aligned}$$

Here,

A is the open loop voltage gain

v_1 is the inverting terminal voltage

v_2 is the noninverting terminal voltage

[Comments \(1\)](#)



Anonymous

v2 is NONinverting

Step 2 of 2

Calculate the voltage input to the inverting terminal of an op-amp.

$$v_o = A(v_2 - v_1)$$

$$\frac{v_o}{A} = v_2 - v_1$$

$$v_1 = v_2 - \frac{v_o}{A}$$

Substitute 150,000 for A , $-40\ \mu\text{V}$ for v_2 , and $15\ \text{V}$ for v_o in the above equation.

$$\begin{aligned} v_1 &= -40 \times 10^{-6} - \frac{15}{150000} \\ &= -40 \times 10^{-6} - 100 \times 10^{-6} \\ &= -140\ \mu\text{V} \end{aligned}$$

Therefore, the input voltage of inverting terminal, v_1 is $\boxed{-140\ \mu\text{V}}$.